CSC_51053_EP, Exercise Session: Recovery in RDBMSs

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1 ACID concept review

1. If we assume disks never crash, do we still need to make an effort to enforce durability? Why?
2. What data structures which are part of the Databases Management System (DBMS) are lost when the server crashes?
3. As a way to ensure durability, how is logging different from replication?
4. What is the utility of checkpointing? What are the advantages and disadvantages of having frequent checkpoints (vs. having them more rarely)? The screen capture below can be used as a hint.

```
[223-235:~/pgData] ioanamanolescu% pg_ctl start -D ~/pgData/
server starting
[223-235:~/pgData] ioanamanolescu% LOG:  database system was shut down at 2019-11-15 09:02:09 CET
LOG:  MultiXact member wraparound protections are now enabled
LOG:  database system is ready to accept connections
LOG:  autovacuum launcher started
LOG:  checkpoints are occurring too frequently (8 seconds apart)
HINT:  Consider increasing the configuration parameter "max_wal_size".
LOG:  checkpoints are occurring too frequently (4 seconds apart)
HINT:  Consider increasing the configuration parameter "max_wal_size".
LOG:  checkpoints are occurring too frequently (2 seconds apart)
HINT:  Consider increasing the configuration parameter "max_wal_size".
LOG:  checkpoints are occurring too frequently (3 seconds apart)
```

5. What tools and algorithms seen in the course enforce:
   (a) Atomicity?
   (b) Consistency?
   (c) Isolation?
   (d) Durability?
6. How does concurrency control impact the *performance* of query (and update) processing? Is the impact positive or negative? Compare:
   - A setting where execution is alway serial (thus, no concurrency, and no concurrency control) with a concurrent setting with concurrency control in place.
   - A normal, concurrent setting with concurrency control, using *"less" concurrency control* vs. *"more" concurrency control*.
7. What are the main aspects of recovery with an impact on performance of query (and update) processing? Is the impact positive or negative?

2 Recovery

1. What are the situations that trigger log entries being flushed to persistent storage?
2. Why does REDO go over log records in the forward order, while UNDO goes over the log records in the reverse order?
3. [1] Consider the schedule in Figure 1. Following the crash, state precisely what is done during: Analysis; Redo; Undo.

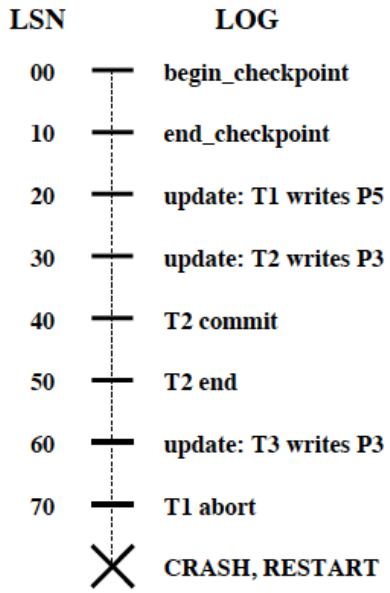


Figure 1: Sample schedule for recovery including a crash.

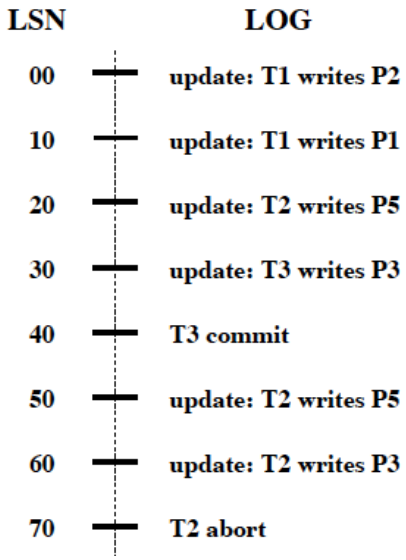


Figure 2: Sample schedule for recovery including an abort of T2.

4. [1] Consider the execution in Figure 2.
   (a) Extend the figure to show prevLSN and undonextLSN values.
   (b) Describe the actions taken to rollback transaction T2.
   (c) Show the log after T2 is rolled back, including all prevLSN and undonextLSN values in log records.

5. Examine the schedule given below. There are three transactions: T_1, T_2, T_3. Initially, the salary is equal to 1 and the tax is equal to 2. Assume WAL and ARIES are used.

	T_1	T_2	T_3
0			START
1			READ tax
2			tax := tax +1
3	START		
4	READ salary		
5	salary := salary +1		
6			WRITE tax
7			COMMIT
8		START	
9		READ tax	
10		READ salary	
11		tax := tax +1	
12		WRITE tax	
13		COMMIT	
14	READ tax		
15	tax := tax + salary		
16	WRITE tax, salary		
17	**System Crashes**		

(a) Show the log entries that would be generated by this execution. Use the same notation used in class. For each log entry, indicate what line above generates it.

(b) When the system is rebooted and tries to recover from the crash which transaction(s) should be undone and which may be redone ?

References

[1] Raghu Ramakrishnan and Johannes Gehrke. *Database management systems (3rd ed.).* McGraw-Hill, 2003.